

A RECAP, WITH TWO WORKED EXAMPLES

Statically indeterminate beams

When equilibrium runs out, compatibility supplies the missing equations

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NOTES

01 · COUNTING

Equilibrium gives three equations; extra supports add unknowns it cannot find

Beam	Reactions r	Equations	Degree $r - 3$
Simply supported (pin + roller)	3	3	0
Propped cantilever (fixed + roller)	4	3	1
Two-span continuous (pin + 2 rollers)	4	3	1
Fixed at both ends	6	3	3

Each redundant needs one **compatibility** equation: a deflection or slope the supports force to be zero.

STATICALLY INDETERMINATE BEAMS

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02 · THE FORCE METHOD

Release one reaction, then make the released beam fit its support again

1. Count the redundants and pick one: here the prop force R_B .
2. Remove it. The primary structure is a cantilever, which is determinate.
3. Find the tip deflection under the load, $\delta_{B,w}$, and per unit R_B , f_{BB} .
4. Compatibility: the real prop does not move, so $\delta_{B,w} - R_B f_{BB} = 0$.
5. Equilibrium now gives every other reaction.

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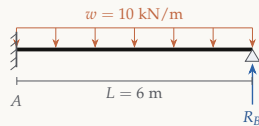
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02 · THE FORCE METHOD

The prop carries three eighths of the load:

$$R_B = 3wL/8$$



$$\delta_{B,w} = \frac{wL^4}{8EI} = 101.25 \text{ mm}$$

$$f_{BB} = \frac{L^3}{3EI}$$

$$R_B = \frac{\delta_{B,w}}{f_{BB}} = \frac{3wL}{8} = 22.5 \text{ kN}$$

Then $R_A = wL - R_B = 37.5 \text{ kN}$ and $M_A = wL^2/8 = 45 \text{ kN m}$, hogging.

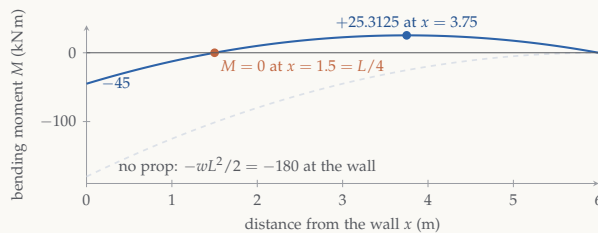
$EI = 16\,000 \text{ kN m}^2$ ($E = 200 \text{ GPa}$, $I = 8 \times 10^{-5} \text{ m}^4$); EI cancels in R_B .

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02 · THE FORCE METHOD

The prop cuts the peak moment from 180 to 45 kN m



Sagging positive: $M(x) = -45 + 37.5x - 10x^2/2 \text{ kN m}$.

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03 · CONTINUOUS BEAMS

The three-moment equation makes the slope continuous over each support

$$M_A L_1 + 2M_B(L_1 + L_2) + M_C L_2 = -\frac{w_1 L_1^3}{4} - \frac{w_2 L_2^3}{4}$$

- Redundants: the support moments. Each span becomes simply supported.
- Each right-hand term is $6EI$ times a simply supported end slope, $wL^3/24EI$.
- Pinned ends: $M_A = M_C = 0$. One interior support, one equation.

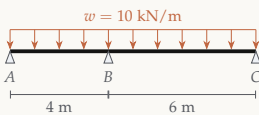
Uniform EI , supports at one level, sagging moments positive (Clapeyron, 1857).

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03 · CONTINUOUS BEAMS

Two spans of 4 and 6 m: the support moment is 35 kN m, hogging



$$\begin{aligned} 2M_B(4 + 6) &= -\frac{10 \cdot 4^3}{4} - \frac{10 \cdot 6^3}{4} \\ 20M_B &= -160 - 540 = -700 \\ M_B &= -35 \text{ kN m} \end{aligned}$$

Span by span: $R_A = 11.25$,
 $R_B = 28.75 + 35.83 = 64.58$, $R_C = 24.17$ kN.

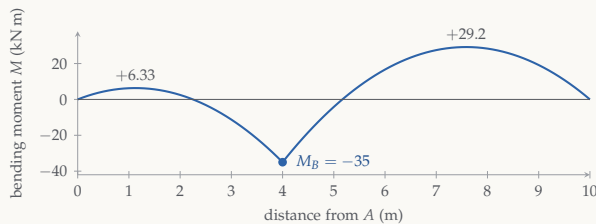
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03 · CONTINUOUS BEAMS

The long span sags most; the support moment is the largest



Sagging positive. Peaks at zero shear: 1.125 m from A, 2.42 m from C.

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04 · CHECKS

Moment distribution reaches the same 35 kN m in one balance

At joint B	member BA	member BC
Stiffness $3EI/L$, share	0.6	0.4
Fixed-end moment $\pm wL^2/8$	+20	-45
Balance the -25 unbalance	+15	+10
Final end moment	+35	-35

End moments are clockwise-positive; ± 35 is the 35 kN m hogging moment found by the three-moment equation.

Far ends pinned: modified stiffness $3EI/L$, no carry-over (Cross, 1930).

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04 · CHECKS

Four checks catch most mistakes before they reach a drawing

- 1. Vertical equilibrium: $11.25 + 64.58 + 24.17 = 100$ kN, and moments about A balance.
- 2. Limit case: equal spans must give $M_B = -wL^2/8$ (45 kN m for $L = 6$ m).
- 3. Compatibility: the deflection comes out zero at every support.
- 4. An independent solver: the site's beam visualizer, by the stiffness method, matches every number here.

Checks 3 and 4: beamdiag's exact-arithmetic reference solver, both examples. 10 / 11

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04 · CHECKS

What to remember

- 1. Count first: the degree of indeterminacy is reactions minus three.
- 2. One compatibility equation per redundant: release it, then make the beam fit again.
- 3. For continuous beams, support moments and the three-moment equation are quickest.
- 4. Check with a second method and with compatibility, not only with equilibrium.

Hibbeler, *Structural Analysis*. Gere & Goodno, *Mechanics of Materials*. Megson, *Structural and Stress Analysis*. Clapeyron (1857), Comptes rendus 45. Cross (1930), *Analysis of continuous frames by distributing fixed-end moments*, Proc. ASCE 56.

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